

Keywords: conformal prediction • decision theory • minimax risk • coverage guarantees

Optimal Decision-Making Based on Prediction Sets

Risk-Optimal Conformal Prediction Derives the Policy and the Set from the Same Minimax Objective

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A conformal prediction set guarantees that the true label falls inside it with probability at least $1 - \alpha$ —but says nothing about what to *do* once you have it. This paper derives the minimax-optimal downstream decision policy for a fixed prediction set, then works backward to the optimal set construction itself, unifying both into a single algorithm: **Risk-Optimal Conformal Prediction (ROCP)**. The result consistently cuts worst-case risk and critical mistake rates relative to standard max-min and best-response baselines, with the largest gains where out-of-set errors are most costly.

AT A GLANCE

Problem: Coverage guarantees say nothing about how to act optimally on a prediction set for a downstream decision.

Why it matters: Prediction sets are increasingly used to drive real decisions (triage, risk limits, control); an uncalibrated decision rule can waste or misuse the coverage guarantee.

Method: Minimax decision theory over distributions consistent with the coverage constraint, solved jointly for the optimal policy and the optimal prediction set.

Result: ROCP reduces worst-case risk and critical mistake rates versus max-min and best-response baselines, especially when out-of-set errors are costly.

Evidence: Comparisons against risk-averse max-min policies and naive best-response policies across decision tasks with varying error costs.

The Big Picture

Conformal prediction has become the standard way to wrap any black-box model with a distribution-free coverage guarantee: the resulting set $C(x)$ contains the true label with probability at least $1 - \alpha$. But in practice, sets are rarely the end product—they feed a downstream decision. A clinician acts on a set of possible diagnoses; a

control system acts on a set of possible next states. Until now, the choice of *how* to act on $C(x)$, and even how $C(x)$ should be shaped for that purpose, has been left to ad hoc heuristics disconnected from the coverage guarantee itself.

This paper treats the full pipeline—set construction and downstream decision—as one decision-theoretic problem, and solves for both pieces jointly under a minimax risk objective.

Why Existing Approaches Fall Short

Two heuristics dominate practice. **Best-response** policies act as if the point prediction were exact, ignoring the coverage guarantee outright—leaving decisions exposed exactly when the true label lands elsewhere in the set. **Max-min** policies plan for the single worst label anywhere in $C(x)$, regardless of how likely the coverage guarantee actually makes it—wasting performance hedging against possibilities the guarantee doesn’t privilege. Neither is derived from the same objective that produced the set in the first place, so neither is calibrated to the decision task.

Core Mechanism

The paper first fixes an arbitrary prediction set $C(x)$ and solves for the minimax-optimal policy: the one minimizing worst-case expected loss over all distributions still consistent with the set’s coverage guarantee. That optimal policy balances the worst-case loss for labels inside $C(x)$ against a penalty for the residual probability mass that could fall outside it. Holding this policy fixed, the authors then solve backward for the prediction set that minimizes the resulting robust risk subject to the coverage constraint—yielding a set shaped for the decision task, not merely for coverage.

Technical Formulation

For decision $a \in \mathcal{A}$ and loss $\ell(a, y)$, define the robust risk of policy π given set C as

$$R(\pi, C) = \sup_{Q: Q(y \in C(x)) \geq 1 - \alpha} \mathbb{E}_Q[\ell(\pi(C(x)), y)].$$

The minimax-optimal π^* balances $\sup_{y \in C(x)} \ell(\pi^*, y)$ against an α -weighted penalty for loss outside $C(x)$. The optimal set $C^*(x)$ then minimizes $R(\pi^*, C)$ subject to $P(y \in C(x)) \geq 1 - \alpha$. ROCP estimates the population quantities needed for π^* and C^* using any black-box probabilistic model, then calibrates a nonconformity threshold on held-out data so that $C^*(x)$ retains exact finite-sample marginal coverage under exchangeability.

Learning or Inference Procedure

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1. Fit a black-box probabilistic model to estimate the population quantities entering the oracle minimax policy π^* .
 2. Construct the candidate set $C^*(x)$ that minimizes the resulting robust risk under the model’s estimates.
 3. Calibrate a nonconformity threshold on a held-out calibration set to enforce exact finite-sample marginal coverage.
 4. At decision time, apply π^* to the calibrated set $C^*(x)$ to produce the action a .

What the Guarantee Says

Regardless of how well the black-box model estimates the oracle quantities, the calibration step guarantees $P(y \in C^*(x)) \geq 1 - \alpha$ in finite samples under exchangeability alone. The risk-optimality of π^* and C^* is with respect to the worst-case distribution consistent with that guarantee—so the decision quality improves as the underlying model improves, while the coverage guarantee never depends on the model being correct.

Experimental Findings

Across decision tasks with varying costs for out-of-set errors, ROCP reduces both worst-case risk and the rate of critical (high-cost) mistakes compared to the max-min baseline (over-conservative) and the best-response baseline (under-protected). The advantage widens as the cost of an out-of-set error grows, precisely the regime where a principled decision policy matters most.

Ablations and Interpretation

Cost sensitivity: When out-of-set costs are low, ROCP, max-min, and best-response converge in performance—the decision-theoretic refinement matters most when mistakes are expensive.

Set size: ROCP’s sets are not simply smaller or larger than standard conformal sets; they are reshaped toward the decision task, sometimes excluding low-cost-if-missed labels in favor of tighter coverage where mistakes are expensive.

Model quality: Decision quality scales with the black-box model’s accuracy, while the coverage guarantee remains exact even under model misspecification—the two guarantees are cleanly separated.

Reference

Tao Wang, Edgar Dobriban. **Optimal Decision-Making Based on Prediction Sets**. arXiv:2602.00989, February 2026. <https://arxiv.org/abs/2602.00989>

Keywords: online conformal prediction • universal portfolio algorithms • regret bounds • adversarial streams

Online Conformal Prediction via Universal Portfolio Algorithms

A Parameter-Free Reduction Turns Coverage Tuning into Portfolio Selection

By Tuo Liu, Edgar Dobriban & Francesco Orabona (University of Pennsylvania, Boston University)

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Online conformal prediction promises long-run coverage even on adversarial data streams—but every prior method needed a hand-tuned learning rate to actually deliver it. This paper shows that controlling a general notion of *linearized regret* implies coverage for *any* online algorithm, then reduces the resulting regret-minimization problem to two-asset portfolio selection. The result, **UP-OCP**, is parameter-free by construction, borrowing decades of tuning-free universal portfolio algorithms to deliver calibrated coverage without a single knob to turn.

AT A GLANCE

Problem: Online conformal methods need a hand-tuned learning rate to guarantee coverage—exactly the kind of choice adversarial streams make impossible to validate in advance.

Why it matters: Mistuned online conformal intervals silently lose their coverage guarantee on the non-stationary or adversarial streams they are meant to handle.

Method: A regret-to-coverage reduction via the Fenchel conjugate, followed by a reduction of the resulting problem to two-asset portfolio selection.

Result: UP-OCP matches or exceeds tuned ACI-style baselines’ coverage with zero learning-rate parameters.

Evidence: Comparisons against Adaptive Conformal Inference (ACI) variants on time-series and adversarial streams.

The Big Picture

Online conformal prediction (OCP) methods maintain a calibrated threshold as data arrive one at a time, aiming for long-run $(1 - \alpha)$ coverage even when the stream is non-stationary or adversarial. The dominant family, descended from Adaptive Conformal Inference (ACI), updates the threshold via an online-gradient-style rule governed by a learning rate. That learning rate trades off how quickly the method tracks distribution shift against how much

it overreacts to noise—and the correct value is stream-dependent and unknowable in advance, precisely in the adversarial setting OCP exists to handle without such foreknowledge.

This paper asks a structural question instead of a tuning question: what property of an online algorithm’s regret actually implies coverage, for any algorithm at all?

Why Existing Approaches Fall Short

ACI-style methods require an algorithm-specific regret analysis for every new update rule, and their coverage guarantees are only as good as the learning rate chosen for that specific rule. Prior refinements made the tuning more forgiving but never removed it. Because the correctness of the guarantee depends on a hyperparameter that cannot be cross-validated on an adversarial stream without assuming away the very adversariality the method is supposed to survive, tuning is not merely inconvenient—it is a genuine gap in the coverage guarantee’s practical reliability.

Core Mechanism

The authors show that controlling *linearized regret* for the $(1 - \alpha)$ -pinball loss implies a long-run coverage bound for any online algorithm, via a black-box reduction that depends only on the Fenchel conjugate of an upper bound on that regret. This decouples “does it cover” from “which algorithm produced the threshold.” Recognizing that the threshold-update problem, viewed through this regret lens, is structurally identical to sequential betting between two assets (covered vs. not covered), they import universal portfolio algorithms—originally designed for sequential investing with no prior knowledge of asset returns—directly into the conformal setting.

Technical Formulation

Let s_t be the nonconformity score at time t and q_t the calibrated threshold, with pinball loss

$$\rho_{1-\alpha}(u) = \begin{cases} (1 - \alpha)u & u \geq 0 \\ -\alpha u & u < 0 \end{cases}, \quad u = q_t - s_t.$$

Define regret

$$R_T = \sum_{t=1}^T \rho_{1-\alpha}(q_t - s_t) - \min_q \sum_{t=1}^T \rho_{1-\alpha}(q - s_t).$$

Empirical coverage error is controlled by R_T through a bound derived from the Fenchel conjugate of the regret’s upper envelope. Casting q_t ’s update as a bet fraction between a “covered” and “not covered” asset reduces the problem to constant-rebalanced portfolio selection, for which universal portfolio algorithms guar-

antee $O(\log T)$ regret with no learning-rate parameter—translating, through the regret-to-coverage bound, into parameter-free coverage.

Learning or Inference Procedure

1. Observe nonconformity score s_t at each round t .
2. Map the threshold-selection problem to a bet fraction over a two-asset (covered / not covered) portfolio.
3. Update the bet using a universal portfolio algorithm—no learning rate is set or tuned.
4. Convert the resulting bet fraction back into the calibrated threshold q_{t+1} and output the prediction interval.

What the Guarantee Says

For any sequence of scores, including adversarially chosen ones, the empirical long-run miscoverage rate deviates from α by an amount controlled by $R_T/T \rightarrow 0$, where R_T is the universal portfolio algorithm’s $O(\log T)$ regret. The bound holds without assuming any distributional structure on the stream and without any user-chosen learning rate.

Experimental Findings

On time-series and deliberately adversarial data streams, UP-OCP matches or exceeds the empirical coverage of ACI variants that required per-stream learning-rate tuning, while UP-OCP itself uses none—removing an entire axis of hyperparameter search from online conformal deployment.

Ablations and Interpretation

Against best-tuned ACI: Even when ACI’s learning rate is tuned in hindsight on each specific stream, UP-OCP’s parameter-free coverage remains competitive, showing little practical cost to removing the tuning step.

Against mistuned ACI: When ACI’s learning rate is chosen without stream-specific tuning, its coverage degrades substantially; UP-OCP’s guarantee is unaffected, since it never depended on a tuned parameter in the first place.

Beyond intervals: The authors note the regret-to-coverage reduction is not intrinsically tied to the two-asset case, suggesting the portfolio-algorithm connection may extend to richer conformal prediction settings.

Reference

Tuo Liu, Edgar Dobriban, Francesco Orabona. **Online Conformal Prediction via Universal Portfolio Algorithms.** arXiv:2602.03168, February 2026.
<https://arxiv.org/abs/2602.03168>

Keywords: conformal risk control • generative models • large language models • memorization

Conf-Gen: Conformal Uncertainty Quantification for Generative Models

Extending Formal Risk Guarantees Beyond Supervised Prediction into LLMs, Image Generators, and Agents

By Gabriel Loaiza-Ganem, Kevin Zhang, Wei Cui, Marc Law & Kin Kwan Leung (Layer 6 AI, TD Bank Group)

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Conformal risk control (CRC) gives formal uncertainty guarantees for supervised predictors—but a diffusion sample or an LLM’s answer has no ground-truth label to calibrate against. **Conf-Gen** relaxes CRC’s assumptions just enough to reach generative systems directly, producing certified guarantees such as “this image is not a memorized copy,” “this dialogue system has asked enough clarifying questions,” and “this agent’s output is correct”—unifying and generalizing prior one-off attempts to bring conformal guarantees to LLMs.

AT A GLANCE

Problem: Conformal risk control assumes labeled prediction pairs; generative and agentic systems have no natural label to calibrate against.

Why it matters: As generative and agentic systems displace classical supervised predictors, the uncertainty-quantification toolkit built for supervised learning risks becoming obsolete.

Method: Relax CRC’s exchangeable-labeled-pair requirement to an exchangeable-generation-instance requirement, calibrating task-specific risk functionals directly on generated outputs.

Result: A single calibration recipe transfers across image generation, dialogue, and agentic tasks without redesign.

Evidence: Demonstrations on non-memorization guarantees for image generators, clarification-sufficiency guarantees for dialogue systems, and correctness guarantees for agent outputs.

The Big Picture

Conformal prediction and its extension, conformal risk control, are the most practically trusted formal uncertainty guarantees in machine learning—but both were built for supervised prediction, where a labeled calibration pair

(x, y) is available. Modern AI progress is increasingly driven by unsupervised generative models: LLMs, image and video generators, and autonomous agents, none of which produce a prediction against a classical ground-truth label. Conf-Gen asks which of CRC’s assumptions are actually load-bearing, and shows that relaxing just the labeled-pair requirement is enough to bring the same finite-sample guarantee machinery to generative tasks.

Why Existing Approaches Fall Short

Prior attempts to bring conformal guarantees to LLMs typically bolted a supervised-style calibration step onto a generative pipeline in a task-specific, one-off way—for instance treating a single decoding step as if it were a classification label. These approaches do not generalize: each new generative task (image synthesis, dialogue, agentic tool use) required its own bespoke reformulation, with no shared theoretical foundation guaranteeing correctness across tasks.

Core Mechanism

Conf-Gen replaces the labeled-pair requirement with task-specific risk functionals defined directly over generated outputs—distance to the training set as a memorization score, a count of clarifying questions asked, or a verifier’s judgment as a correctness score. Calibration proceeds exactly as in CRC: find the smallest threshold for which the empirical risk, corrected for finite-sample slack, does not exceed the target level. What changes is the object being calibrated: a single generative process’s own outputs, rather than a labeled prediction task.

Technical Formulation

Let $R(\lambda) = \mathbb{E}[\ell(\lambda, X)]$ be the population risk as a function of threshold λ , for a monotone bounded loss ℓ . Calibration selects

$$\hat{\lambda} = \inf \left\{ \lambda : \hat{R}_n(\lambda) + c_n(\delta) \leq \alpha \right\},$$

where $\hat{R}_n$ is the empirical risk over n calibration instances and $c_n(\delta)$ is a finite-sample correction term ensuring $R(\hat{\lambda}) \leq \alpha$ with probability at least $1 - \delta$. Conf-Gen’s relaxation replaces the requirement that (X, Y) be an exchangeable labeled pair with the weaker requirement that the sequence of *generation instances* (and their associated risk scores) be exchangeable—sufficient to preserve the guarantee while accommodating memorization scores, clarification-sufficiency scores, and agent-correctness scores as the risk functional.

Learning or Inference Procedure

1. Define a task-specific, monotone risk functional over generated outputs (memorization distance, clarification-sufficiency, agent-correctness).

- tion count, verifier score).
2. Collect an exchangeable sequence of generation instances and compute the empirical risk at a grid of thresholds λ .
3. Calibrate $\hat{\lambda}$ as the smallest threshold whose finite-sample-corrected empirical risk meets the target α .
4. At deployment, apply the calibrated threshold to new generation instances, inheriting the risk guarantee.

What the Guarantee Says

For any monotone bounded risk functional and any exchangeable sequence of generation instances, $R(\hat{\lambda}) \leq \alpha$ holds with probability at least $1 - \delta$ in finite samples—matching the guarantee strength of classical CRC, despite the absence of a conventional labeled calibration set.

Experimental Findings

The paper demonstrates Conf-Gen across three qualitatively different generative settings: certifying that image generator outputs are not memorized copies of training data, certifying that a conversational AI has asked sufficient clarifying questions before committing to an answer, and certifying the correctness of an AI agent’s output—showing the same calibration recipe transfers across modalities and task types without redesign.

Ablations and Interpretation

Unification: Conf-Gen recovers prior ad hoc LLM conformal methods as special cases, showing they were instances of a single underlying relaxation rather than unrelated tricks.

Generality: The framework extends to tasks with no natural notion of a held-out label at all, such as certifying properties of an agent’s multi-step trajectory.

Practical takeaway: Task designers need only supply a monotone risk functional over generation instances; the calibration and guarantee machinery is otherwise unchanged from classical CRC.

Reference

Gabriel Loaiza-Ganem, Kevin Zhang, Wei Cui, Marc Law, Kin Kwan Leung. **Conf-Gen: Conformal Uncertainty Quantification for Generative Models**. arXiv:2605.28920, May 2026. <https://arxiv.org/abs/2605.28920>

Keywords: conformal risk control • robust optimization • Pareto frontier • predict-then-optimize

Calibrating Decision Robustness via Inverse Conformal Risk Control

Tracing the Miscalibration-Regret Pareto Frontier Instead of Fixing a Coverage Target in Advance

By Wenbin Zhou & Shixiang Zhu (Carnegie Mellon University)

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Robust optimization protects decisions by hedging against a conformal uncertainty set—but the robustness level behind that set is usually fixed ad hoc, either exposing decisions or wasting performance on needless conservatism. **Inverse Conformal Risk Control (ICRC)** flips the usual recipe: instead of fixing coverage and later measuring regret, it delivers simultaneous, distribution-free, finite-sample guarantees on both miscalibration *and* regret across an entire family of policies, letting decision-makers pick their robustness level directly off a certified Pareto frontier.

AT A GLANCE

Problem: The coverage level behind a robust conformal decision policy is normally fixed a priori, with no statistical way to judge whether it was the right choice.

Why it matters: As conformal sets increasingly feed robust optimization pipelines in finance, operations, and safety-critical planning, an unexamined robustness knob leaves performance on the table invisibly.

Method: Simultaneous, distribution-free finite-sample bounds on both miscalibration and regret across a family of robustness levels, valid uniformly.

Result: A certified miscalibration-regret Pareto frontier that decision-makers can select from *after* seeing it, with the guarantee still holding.

Evidence: Robust predict-then-optimize benchmarks comparing ICRC’s certified frontier against fixed- α conformal robust optimization.

The Big Picture

In robust predict-then-optimize, a conformal uncertainty set $\mathcal{U}_\alpha(x)$ is built at coverage level α , and a decision minimizes the worst-case cost over that set. The choice of α determines both the protection the decision receives and the optimization performance (regret) sacrificed for that protection—but conventional practice fixes α once, in advance, with no principled way to know whether that choice was too cautious or not cautious enough for the task’s

real cost structure. This paper makes the entire tradeoff between protection and performance a statistically certified, inspectable object rather than a single unexamined hyperparameter.

Why Existing Approaches Fall Short

Fixing a single coverage target α and only afterward measuring the regret it induces means the decision-maker learns whether their choice was good only after committing to it. Re-running the pipeline for a different α offers no statistical guarantee that the new choice is valid, since the original calibration was only certified for the original α . This traps practitioners into a single ad hoc choice with no honest way to compare alternatives.

Core Mechanism

Rather than treating α as a knob set once, ICRC treats the entire curve relating miscalibration to regret as the object of interest. It constructs a family of policies indexed by robustness level and provides distribution-free, finite-sample bounds on both the true miscalibration rate and the true regret at every level in that family *simultaneously*, so a decision-maker can inspect the empirical Pareto frontier after the fact and select whichever point matches their actual cost-risk preference—with the guarantee still holding for that a posteriori choice.

Technical Formulation

For a family of robust policies $\{\pi_\alpha\}$ built from uncertainty sets $\mathcal{U}_\alpha(x)$, define

$$\text{miscalibration}(\alpha) = P(y \notin \mathcal{U}_\alpha(x)),$$

$$\text{regret}(\alpha) = \mathbb{E}[\text{cost}(\pi_\alpha(x), y)] - \min_a \mathbb{E}[\text{cost}(a, y)].$$

These move in opposite directions as α varies, forming a Pareto frontier. ICRC constructs simultaneous confidence bands for both functions, valid uniformly over a grid of α values with probability at least $1 - \delta$, using conformal calibration on a held-out set combined with uniform concentration arguments (in the style of DKW-type bounds) to control the multiple-testing-like inflation from evaluating many α values on the same data.

Learning or Inference Procedure

1. Build a grid of candidate robustness levels α and their corresponding uncertainty sets $\mathcal{U}_\alpha(x)$.
2. On held-out calibration data, estimate miscalibration and regret for every α in the grid.
3. Apply uniform concentration bounds to certify simultaneous validity of both estimates across the whole grid.

4. Present the certified empirical Pareto frontier; the decision-maker selects $\hat{\alpha}$ off the frontier according to their cost-risk preference.

What the Guarantee Says

With probability at least $1 - \delta$, the true miscoverage and true regret curves are simultaneously bracketed by the certified bands at every α on the grid—so whichever $\hat{\alpha}$ the decision-maker picks *after* inspecting the frontier still enjoys valid finite-sample guarantees on both quantities, unlike a guarantee calibrated for one α chosen in advance.

Experimental Findings

On robust predict-then-optimize benchmarks, ICRC’s certified Pareto frontier lets practitioners identify robustness levels that achieve materially lower regret than conventional fixed- α conformal robust optimization at matched miscoverage, and vice versa—revealing that ad hoc α choices in prior pipelines were leaving performance on the table without practitioners ever being able to see it.

Ablations and Interpretation

Grid resolution: Finer grids over α tighten the achievable frontier at a modest cost in the width of the simultaneous confidence bands, reflecting the usual multiple-testing tradeoff.

Comparison to fixed- α baselines: Any single fixed- α policy corresponds to one point on ICRC’s frontier; the benefit is purely in being able to see and choose among the alternatives with statistical validity.

Practical takeaway: ICRC does not require the analyst to predict the right robustness level in advance—it makes the tradeoff observable and lets the choice come last.

Reference

Wenbin Zhou, Shixiang Zhu. **Calibrating Decision Robustness via Inverse Conformal Risk Control**. arXiv:2510.07750, October 2025. <https://arxiv.org/abs/2510.07750>

Keywords: conformal prediction • label efficiency • sequential stopping rules • calibration cost

Provably Label-Efficient Conformal Prediction

A Sequential Stopping Rule Cuts Calibration Labeling Cost by 41% Without Sacrificing Coverage

By Andrew Ilyas, Joonhyuk Ko, Jingwu Tang, Zhiwei Steven Wu & Jiahao Zhang (Massachusetts Institute of Technology, Carnegie Mellon University)

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Conformal prediction’s finite-sample coverage guarantee is usually presented as free once a calibration set exists—but collecting those labels rarely is. This paper treats calibration labels as a costly, queryable resource and designs an online stopping rule that automatically balances labeling cost against prediction-set efficiency. The rule matches the best *fixed* calibration size chosen with hindsight, cuts total labeling cost by **41.4% ± 2.3%**, and provably preserves exact coverage despite deciding, on the fly, when to stop labeling.

AT A GLANCE

Problem: Conformal calibration-set size is treated as a free given, but labels needed to build it are often expensive to collect.

Why it matters: Medical labeling, expert annotation, and expensive simulation make calibration cost the dominant cost of deploying conformal prediction in practice.

Method: An online, cost-aware stopping rule for sequential label queries that halts once further labels are provably not worth their cost.

Result: Matches the best fixed calibration size in hindsight; cuts total labeling cost by 41.4% ± 2.3% empirically.

Evidence: Benchmark classification and regression tasks comparing adaptive stopping against fixed calibration-set-size baselines, with coverage validated throughout.

The Big Picture

Split conformal prediction fixes a calibration set of size n up front; more calibration data generally tightens prediction sets, with diminishing returns, and every additional label carries a real acquisition cost. Conformal theory has always treated n as given, leaving practitioners to guess a calibration budget with no statistical guidance.

This paper asks the practical question directly: given unlabeled examples arriving one at a time and the ability to query a label at a cost, when should calibration stop and prediction begin?

Why Existing Approaches Fall Short

Practitioners today either guess a calibration-set size in advance, risking wasted labeling budget or an under-calibrated, oversized prediction set, or they collect labels until budget runs out with no statistical link between that stopping point and the tradeoff they actually care about. Naively stopping “when the sets look tight enough” is an adaptive, data-dependent rule that would ordinarily void the exchangeability conformal prediction relies on—so ad hoc adaptive stopping has been avoided rather than solved.

Core Mechanism

The authors formalize a total-cost objective combining the cumulative cost of labels queried so far with the expected inefficiency (prediction-set size) of the conformal predictor that would result from stopping now. They design a sequential stopping rule that halts querying once the estimated marginal benefit of one more label—in expected set-size reduction—no longer exceeds its cost. The delicate technical contribution is showing that this data-dependent stopping decision can be constructed so the resulting calibration set still yields a conformal predictor with exact finite-sample coverage, regardless of the adaptive stopping.

Technical Formulation

With per-label cost c and a weight λ trading off cost against expected prediction-set size $\mathbb{E}[|C(X)|]$, define the total objective after m queried labels as

$$J(m) = m \cdot c + \lambda \cdot \mathbb{E}[|C_m(X)|],$$

where C_m is the conformal predictor calibrated on the first m labels. The stopping time is

$$\tau = \inf\{m : \hat{\Delta}(m) \leq 0\},$$

where $\hat{\Delta}(m)$ estimates the marginal value of querying label $m + 1$. Under mild regularity conditions, $\mathbb{E}[J(\tau)]$ matches $\min_m J(m)$ —the best fixed calibration size chosen with hindsight—up to lower-order terms, while $P(Y \in C_\tau(X)) \geq 1 - \alpha$ holds exactly despite τ being a data-dependent stopping time.

Learning or Inference Procedure

1. Observe an unlabeled example; decide whether to query its label based on the current estimate of marginal value $\hat{\Delta}(m)$.

2. If $\hat{\Delta}(m) > 0$, query the label, add it to the calibration set, and increment m .
3. If $\hat{\Delta}(m) \leq 0$, stop querying at $\tau = m$.
4. Calibrate the conformal predictor on the τ collected labels and deploy it with the standard coverage guarantee.

What the Guarantee Says

Despite τ being chosen adaptively from the data observed so far, the resulting conformal predictor satisfies $P(Y \in C_\tau(X)) \geq 1 - \alpha$ exactly, in finite samples. Separately, the expected total cost at the stopping time matches the best achievable cost under a fixed calibration budget selected with hindsight knowledge of the data.

Experimental Findings

Across benchmark classification and regression tasks, the proposed stopping rule reduces total labeling cost by **41.4% $\pm$ 2.3%** relative to fixed calibration-set-size baselines chosen without this guidance, while maintaining nominal coverage throughout the evaluated tasks.

Ablations and Interpretation

Cost sensitivity: As per-label cost c increases relative to the set-size weight λ , the stopping rule halts earlier, trading a somewhat larger prediction set for lower total cost—exactly the intended behavior.

Robustness to misestimation: Errors in estimating $\hat{\Delta}(m)$ affect the total-cost optimality of the stopping point but never the coverage guarantee, which is preserved by construction regardless of how τ is chosen.

Practical takeaway: Calibration-set size, long treated as a free design choice in conformal prediction, can instead be solved for directly as a cost-aware stopping problem.

Reference

Andrew Ilyas, Joonhyuk Ko, Jingwu Tang, Zhiwei Steven Wu, Jiahao Zhang. **Provably Label-Efficient Conformal Prediction**. ICML 2026. <https://icml.cc/virtual/2026/poster/65796>